

# Esophagectomy

## 1 Introduction. L 1

I'm Dr Jonathan Salo, a GI Cancer Surgeon in Charlotte, North Carolina. These videos are designed to educate you about cancer and its treatment and help you and your cancer care team make the right decisions for you.

Of course, there is no substitute for the expert opinions of your cancer care team.

Surgery of the Stomach 7 (2)

In this video we'll discuss *surgery of the stomach* and we will focus on

- Different Types of Operations
- Impact that these operations have on quality of life after surgery

*Gastrectomy* is the surgical removal of the stomach (3)

... and for some tumors this means removal of lymph nodes as well (4)

## 2 How much Stomach? (5)

The differences between operations on the stomach is how much stomach is removed and how much remains.

This is dependent upon:

- The tumor Type
- The tumor location

## 3 Types of stomach Tumors (6)

Another term is esophagogastrectomy (4)

## 4 Esophagectomy

Esophagectomy is generally performed for three different situations:

- Superficial Tumors (T1) that can't be completely removed by endoscopy (5)

Superficial tumors are T1. If this terminology is not familiar to you, this may be a good time to view our video about Esophageal Cancer Staging and Treatment

We also have a separate video about superficial or “early stage” tumors and how they can sometimes be removed through endoscopic therapy without the need for surgery.

## 5 Esophagectomy.

Esophagectomy is also performed for

- Localized Tumors that are T1b or T2 (6)

## 6 Esophagectomy

- Locally Advanced Tumors, which are T3 or Node-positive (6)

## 7 Esophagectomy

Locally-advanced tumors usually treated with some sort of preoperative therapy prior to surgery.

Preoperative therapy is usually some combination of chemotherapy, immunotherapy or radiation therapy

We have videos which discuss locally-advanced esophageal cancer, which you may find helpful.

## 8 Goals of Esophagectomy

The goals of esophagectomy are to remove the tumor (9)

## 9 Goals of Esophagectomy

And nearby lymph nodes (10)

## 10 Goals of Esophagectomy

An *additional* portion of the esophagus and stomach is also removed with the tumor to reduce the risk of recurrence (12)

## 11 Reconstruction

In most cases, the lower part of the stomach is fashioned into a tube, to create a new esophagus.  
## Reconstruction

The esophagus and stomach tube are now joined together

The new connection between esophagus and stomach is called the anastomosis (14)

## 12 Types of Esophagectomy. 4

There are two major types of esophagectomy:

Partial esophagectomy removes the lower 2/3 of the esophagus

Total esophagectomy removes all of the esophagus (15)

## 13 Partial Esophagectomy 2

During a partial esophagectomy, the new connection between esophagus and stomach (or anastomosis) made in the chest (16)

## 14 Ivor Lewis Esophagectomy

A partial esophagectomy is also referred to as an Ivor Lewis esophagectomy

## **15 Total Esophagectomy**

A total esophagectomy removes all of esophagus, and a new connection (or anastomosis) made in the neck (18)

## **16 Total Esophagectomy**

Two different approaches to total esophagectomy

Transhiatal approach or

McKeown which is also known as the three-incision esophagectomy (19)

## **17 Which Esophagectomy? 4**

Your surgeon will recommend a surgical approach based upon a number of factors.

One factor is the location of the tumor (20)

## **18 Distal - Partial 2**

For tumors located in the lower, or distal part of the esophagus, a Partial esophagectomy will remove the cancer and enough additional esophagus to ensure that the cancer is completely removed (21)

## **19 Distal - Partial**

So for distal cancers, an Ivor Lewis Approach is a common operation (22)

## **20 Upper = Proximal**

For tumors located in the upper or proximal part of the esophagus, a total esophagectomy is required to remove all of the cancer.

## 21 Which Esophagectomy - Surgeon 4

Another important factor is surgeon preference and experience.

You will have an opportunity to discuss the operation when you meet with your surgeon

## 22 Why Remove so much Esophagus?

A question I am commonly asked is “why do you need to remove so much esophagus?=”

## 23 Esophagectomy

On the left is a diagram of a standard Ivor Lewis esophagectomy.

On the right is a proximal gastrectomy. The operations are different because much *less* esophagus is removed with a proximal gastrectomy.

With a proximal gastrectomy, the anastomosis is much lower in the chest cavity.

## 24 Ivor Lewis Esophagectomy vs Proximal

The Ivor Lewis approach removes *more* esophagus and the proximal gastrectomy removes *less*

On the right is a proximal gastrectomy. The operations are different because much *less* esophagus is removed with a proximal gastrectomy.

With a proximal gastrectomy, the anastomosis is much lower in the chest cavity.

## 25 Ivor Lewis Esophagectomy vs Proximal

As it turns out, there is *more* acid reflux with a proximal gastrectomy.

## 26 Ivor Lewis Esophagectomy vs Proximal

We have found that the quality of life is better if about half to two-thirds of the esophagus is removed, with the result that the anastomosis is a bit higher in the chest.

## **27 Surgical Techniques 2**

Another aspect of surgery are the techniques used.

An esophagectomy can be performed with open techniques or minimally-invasive techniques (25)

## **28 Open Surgery**

An open esophagectomy uses a laparotomy incision to access the abdomen, and a thoracotomy to access the chest (26)

## **29 Minimally InVasive**

Minimally-invasive technique uses small incisions and specialized instruments.(27)

## **30 Minimally InVasive**

These techniques can reduce discomfort and length of hospital stay. (28)

## **31 Laparoscopic techniques**

Laparoscopic techniques use special instruments to work through small incisions. These are directly controlled by the surgeon. (28)

## **32 Robotic techniques**

In some cases, robotic techniques can be used. The surgeon controls the robotic instruments from a console (29) which allows 3-D view inside the abdominal cavity

## **33 Robotic techniques**

The surgeon operates the hand controls (30)

### **34 Robotic techniques**

...and is able to direct the movements of the robotic arms (31)

### **35 Robotic techniques**

The robot arms are attached to instruments which which enter the abdomen through small incisions (32)

### **36 Robotic techniques**

The robot precisely mimics the actions of the surgeon (33)

If you search on the internet for “surgical robot grape” you can see the robot in action as it is used to suture the skin back on a grape.

You will get a feel for the precision of a surgical robot.

### **37 Risks of Surgery 4**

There are complications of surgery you will want to discuss with your surgeon

- Leakage from the anastomosis
- Pneumonia

### **38 Leak from Anastomosis 2**

The anastomosis is the new connection made between the esophagus and the stomach.

If the anastomosis does not heal properly. (34)

### **39 Leak from Anastomosis**

There will be leakage of fluid from inside the esophagus into the chest cavity. (3=5)

## **40 Leak from Anastomosis**

This can lead to infection (36)

## **41 Leak from Anastomosis**

and a prolonged hospital stay (37)

## **42 Treatment of Leak**

In some cases, small leaks will heal as long as there is adequate nutrition (38)

## **43 Stent**

IN other cases a stent is placed inside the esophagus to patch the leak from the inside (40)

## **44 Esophageal Stent**

The stent is a wire mesh tube covered in plastic, which is placed inside the esophagus by endoscopy. (41)

## **45 Esophageal Stent**

Once healing has occurred, the stent is removed, typically about 6 weeks later (42)

## **46 Pneumonia 2**

Pneumonia is an infection in the lungs(48)

which come from bacteria in the mouth



## **47 Secretions from the Mouth**

The secretions in the mouth contain bacteria (50)  
and ordinarily these secretions pass into the esophagus  
and are swallowed

## **48 Secretions from the Mouth**

Those secretions *can* occasionally enter the windpipe (53)

## **49 Secretions from the Mouth**

But are ordinarily quickly cleared by a cough (54)

## **50 Secretions in the Airway**

After surgery... particularly after patients have been asleep for a period of time  
Secretions in the airway more easily pass into the lungs  
...and can cause pneumonia

## **51 Pneumonia Prevention after Surgery**

Pneumonia prevention after surgery means clearing secretions from the lungs by Coughing  
(61)

## **52 Pneumonia Prevention after Surgery**

Deep breathing (62)

## **53 Pneumonia Prevention after Surgery**

and Walking (63)

## **54 INcentive spirometer**

An incentive spirometer is a device which can help coach you to take deep breaths after surgery (64)

## **55 Pain control**

Coughing, deep breathing, and walking after surgery can be uncomfortable, so another important strategy to prevent pneumonia is to work to minimize discomfort. (67)

## **56 Pain control**

Minimally-invasive techniques, either laparoscopic or robotic, can help reduce discomfort by the use of small incisions (66)

## **57 Pain control**

The other strategy is pain control

## **58 Epidural Catheter**

An epidural catheter can help with postoperative pain control.

This is similar to the epidural used for childbirth.

A small tube is placed under the skin near the spine with local anesthetic (68)

## **59 Epidural Catheter**

This allows the administration of pain medicines which can provide pain relief without sedation (69)

This makes it easier to walk without discomfort.

## **60 Epidural Catheter**

The goal is to minimize pain (70)

And make it easier to cough, deep breathe, and walk after surgery

## **61 1**

We hope you have found this video helpful. Here are links to some other videos.

Feel free to leave a comment or a question, or if you have suggestions for future videos.

If you or a family member have had an encounter with esophagectomy surgery, I would love to hear about your experience, so please take a minute to leave a comment below.

We're constantly creating new videos, so please subscribe to be notified of new videos when we post them. (29)